

Jessica Imlau Dagostini

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🏠 Personal Page in LinkedIn 🐙 GitHub

AREAS OF INTEREST

High-Performance Computing | Optimization | Scientific Computing | Load Balancing

EDUCATION

- Sep, 2022 – Present **Ph.D. Student** in Computer Science
University of California Santa Cruz (UCSC), Santa Cruz, United States
Expected Graduation Date: June, 2028
Advisor: Prof. Dr. Abel Souza
- Mar, 2020 – Mar, 2022 **M.Sc.** in Computer Science (GPA: A)
Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil
"Performance Improvements Applied in an Electromagnetic Inversion Application Focused on Homogeneous and Heterogeneous Computational Environments"
Advisor: Prof. Dr. Lucas Mello Schnorr
- Feb, 2015 – Dec, 2019 **B.Sc.** in Computer Science (Summa Cum Laude – 9.39 / 10)
Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil
"Evaluating the Container's Applicability in a Distributed System of Code Judgement"
(Avaliação da Aplicabilidade de Containers em um Sistema Distribuído de Julgamento de Códigos.)
Advisor: Prof. Ms. Neilor Avelino Tonin
Co-advisor: Dr. Jean Luca Bez

HONORS & FELLOWSHIPS

- Aug, 2024 **2024 ACM SIGHPC Computational and Data Science Fellowship**
Highly competitive fellowship sponsored by ACM SIGHPC focused on underrepresented groups working with High Performance Computing. 2024 cohort composed only by two fellows.
- Apr, 2024 **CRA-WP Grad Cohort for Women**
Selected student to participate in the annual Grad Cohort for Women on Minneapolis, MN.
- 2023 **2^o Place - ISC Virtual Student Cluster Competition**
Team Not-So-Slow-Slugs - UCSC. Awarded by ISC High Performance 2023 Conference.
- Apr, 2023 – Jun, 2023 **UCSC Dean's Fellowship**
Academic tuition coverage given to a small number of outstanding admitted PhD students at UCSC.
- 2021 **Students@SC'21 Best Lightning Talk**
Awarded by Students@SC Program from Supercomputing'21 Conference
- Mar, 2020 – Mar, 2022 **Master Scholarship**
Financial scholarship provided by Petróleo Brasileiro, to work in the research project "Otimização Computacional de Códigos de Inversão de Dados" (Computational Optimization of Data Inversion Codes)
- 2020 **URI Erechim Best Student Award** SUMMA CUM LAUDE (9.39 / 10.0)
Awarded by Universidade Regional Integrada do Alto Uruguai e das Missões (URI)
- 2020 **SBC Highlight Student Award** SUMMA CUM LAUDE (9.39 / 10.0)
Awarded by Brazilian Computer Society (SBC)
- 2016 **2^o Place - Tecnomate Contest 2016, Categoria "Libres"**
Awarded by Departamento de Ingeniería en Sistemas de Información, UTN Santa Fé (Argentina).
- Fev, 2015 – Dez, 2019 **University for All Program (ProUni)**
Financial and merit-based full scholarship provided by Brazilian Federal Government to cover university tuition (selected as 1st place)
- 2010 **Honorable Mention on 6th Brazilian Math Olympiads of Public Schools (OBMEP)**
Awarded by Conselho Nacional de Pesquisa (CNPq), Brasil.

PROFESSIONAL EXPERIENCE

Jan, 2023 – Ongoing

Teaching Assistant

University of California, Santa Cruz, Santa Cruz, United States

Courses:

- CSE113 - Parallel and Concurrent Programming (Winter 2023, Winter 2024, Fall 2024, and Winter 2026)

June, 2025 – Sept, 2025

Graduate Student Research Assistant Intern

National Energy Research Scientific Computing Center (NERSC)

Lawrence Berkeley National Lab

Mentor: Dr. Kevin Gott

- Development of a toolbox to extract load balancing metrics from real simulation from WarpX, a particle-based application that utilizes AMReX, allowing the easy testing of different load balancing strategies
- Implementation of support of memory capacity as a constraint in the load balancing algorithms (with modifications in both AMReX and WarpX source-code)

Sept, 2022 – Ongoing

Graduate Student Researcher

University of California, Santa Cruz, Santa Cruz, United States

Co-advisor: Prof. Dr. Scott Beamer

Co-advisor: Prof. Dr. Tyler Sorensen

Co-advisor: Prof. Dr. Abel Souza

- Development of a proxy application based on this pangenome mapping tool to allow the exploration of software improvements
- Understanding the impact of load balancing algorithms on power-efficiency usage on distributed systems

Sept, 2023 – Dec, 2023

PhD Intern – High Performance Computing

Pacific Northwest National Laboratory, Richmond, United States

Advisor: Dr. Joseph B Manzano

- Performance characterization of a DNA mapping application that uses pangenomes as references and a graph structure called GBWT (Graph Burrows-Wheeler Transform) to index DNA readings
- Started working towards developing a proxy application based on this pangenome mapping tool

Jan, 2021 – Jul, 2022

Principal System Architect

beecrowd, São Paulo, Brazil

- Responsible for the development and maintenance of the programming language support and setup of the beecrowd judging system;
- Organize and supervise team's activities.

Mar, 2020 – Mar, 2022

Graduate Student Researcher

Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil

"Computational Optimization of Data Inversion Codes"

(Otimização Computacional de Códigos de Inversão de Dados)

Advisor: Prof. Dr. Lucas Mello Schnorr

- This project investigated and implemented computational techniques that improved the execution and quality of results of Data Inversion codes;
- Evaluated, proposed and applied algorithms to improve the load balancing and scalability of a data inversion application of survey oil and gas in homogeneous and heterogeneous systems.

May, 2019 – Sep, 2019

Adjunct Faculty

National Service for Commercial Education - RS, SENAC/RS, Erechim, Brasil.

- Taught robotics to K-12 students in a project complementary to school;
- Teacher in two components of the Fundamentals of Informatics course (Compact Internet and Compact PowerPoint).

Jan,2019 – Mar,2019

Intern Researcher (Summer Program)

Brazilian Center for Research in Energy and Materials (CNPEM), Campinas, Brazil

Brazilian Synchrotron Light Laboratory (LNLS), Brasil

*"GPU Code Implementation to Phase Recovery on Fraunhofer Region"**(Implementação em GPU para a Recuperação da Fase na Região de Fraunhofer)*

Advisor: Dr. Eduardo Xavier Miqueles

- Summer internship project consisted of the development of a specific tool for the restoration of projective images obtained in the Fraunhofer region;
- Main developer of an application in CUDA to be incorporated into the pipeline of the automatic processing of the detectors from the Sirius synchrotron light source;
- Discovered some bugs at cuFFT library that were reported to NVIDIA developers.

Aug,2018 – Jul,2019

Undergraduate Student Extensionist

Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil

*"Insertion of High School Students from Erechim on Brazilian Olympiads in Informatics"**(A Inserção de Alunos do Ensino Médio de Erechim na Olimpíada Brasileira de Informática)*

Advisor: Prof. Ms. Neilor Avelino Tonin

- Project to teach algorithms and programming languages to High School students
- Taught programming to 45 students that enrolled in the Brazilian Olympiads in Informatics (OBI)

Aug,2015 – Jul,2018

Undergraduate Research Assistant

Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil

- *"A Study About Distributed Systems to be Applied on URI Online Judge Portal"*
(Um Estudo Sobre Sistemas Distribuídos para Aplicação no Portal URI Online Judge)

Advisor: Prof. Ms. Neilor Avelino Tonin

- Distributed the front-end servers from URI Online Judge;
- Studied and applied tests to verify which system distribution will be the best option for the environment

- *"Developing a Customized Online Forum to URI Online Judge"*
(Criação de um Fórum Customizado para o Portal URI Online Judge)

Advisor: Prof. Ms. Neilor Avelino Tonin

- Rebuilt the online forum from the URI Online Judge (now beecrowd) platform, applying some concepts related to Human-Computer Interaction, developed using CakePHP framework and PostgreSQL database;
- The new version was deployed in April, 2017, now being used by 300k users, with more than 8800 posts (<https://www.becrowd.com.br/judge/en/forum>).

Apr,2012 – Aug,2014

Junior Student Research

Universidade Regional Integrada do Alto Uruguai e das Missões - URI Erechim, Erechim, Brazil

Escola Estadual Normal José Bonifácio (High School), Brazil

*"Discussion and Deepening of Concepts Regarding the Study of Mathematical Functions on High School"**(Discussão e Aprofundamento de Conceitos a Respeito do Estudo de Funções do Ensino Médio)*

Advisor: Prof. Ms. Clemerson Alberi Pedrosa

- Investigated, discuss, and deepen mathematical concepts related to the study of functions and contribute to the approximation of scientific-theoretical knowledge didactic-operational knowledge in Exact Sciences;
- Developed a diversity of studies from applied mathematics and created a collection of problems modeled from real-world problems.

PUBLICATIONS**FULL PAPERS PUBLISHED IN CONFERENCES**

- **Dagostini, J.;** Manzano, J., Sorensen, T., Beamer, S. **miniGiraffe: a Pangenome Proxy App.**
In: *2025 IEEE International Symposium on Workload Characterization (IISWC)*, Irvine, CA, USA, 2025, pp. 339-352, <https://doi.org/10.1109/IISWC66894.2025.00036>.

- **Dagostini, J.**; H. C. P. da Silva, V. G. Pinto, R. M. Velho, E. S. L. Gastal and L. M. Schnorr. (2021) **Improving Workload Balance of a Marine CSEM Inversion Application**. In: *2021 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, 2021, pp. 704-713, <https://doi.org/10.1109/IPDPSW52791.2021.00107>.
- **Dagostini, J.**; Lima, M. V. de M.; Bez, J. L.; Tonin, N. A. (2018) **URI Online Judge Blocks: Construindo Soluções em uma Plataforma Online de Programação**. In: *XXIX Simpósio Brasileiro de Informática na Educação (Brazilian Symposium on Computers in Education)*, Fortaleza, 2018. v. 1. p. 168-177. <http://dx.doi.org/10.5753/cbie.sbie.2018.168>

SHORT PAPERS AND POSTERS PUBLISHED IN CONFERENCES

- Yellapragada, S., **Dagostini, J.**, Gott, K., Hartman-Baker, R. (2025) **Heterogeneity-Aware Task Allocation for Modern HPC Systems**. In *SC25: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, St. Louis, MO, USA. **Best Research Poster Finalist**.
- **Dagostini, J.**, Yellapragada, S., Gott, K., Hartman-Baker, R. (2025) **1** In *SC25: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, St. Louis, MO, USA.
- **Dagostini, J.**; Pinto, V., & Schnorr, L. (2021). **Aplicação de Método Guloso para Balanceamento de Carga em uma Aplicação de Exploração Eletromagnética**. In: *Anais da XXI Escola Regional de Alto Desempenho da Região Sul*, (pp. 87-88). Porto Alegre: SBC. <https://doi.org/10.5753/erads.2021.14782>
- **Dagostini, J.**; Lima, M. V. de M.; Bucior, L.; Bez, J. L.; Tonin, N. A. (2017) **Incentivando a Aprendizagem de Algoritmos Através do URI Online Judge Forum 2.0** In: *XXVIII Simpósio Brasileiro de Informática na Educação (Brazilian Symposium on Computers in Education)*, Recife, 2017. v. 1. p. 1781. <http://dx.doi.org/10.5753/cbie.sbie.2017.1781>

PRESENTATIONS AND INVITED TALKS

INVITED TALKS

1. **Dagostini, J.**; Beamer, S., & Sorensen, T., & Manzano, J., & Marques, A. (2024). **Developing a Proxy Application Based on a Parallel Pangenome Mapping Tool**. In: *The 1st Workshop on Emerging Computer Systems Challenges and Applications in Biomedicine (BioSys)*. San Diego, CA, USA.
2. **Dagostini, J.**; Solórzano, A. L.; Girelli, V.; Carmin, M. (2022) **Special Sessions - Women in High Performance Computing**. (WHPC Affiliate Brasil - Região Sul. XXII Escola Regional de Alto Desempenho, Parana, Brasil
3. **Dagostini, J.**; Gayler, R.; Beste E. de;F ratkin, M.; Merritt, C.; Gonsiorowski, E.; Belcher, K.; Chen, D. (2021) **Addressing Challenges in HPC Careers**. *Students@SC Student Programming, Supercomputing 21 Conference, St. Louis, United States*
4. **Dagostini, J.**; Baptista, V.; Zhan, Y.; Leiva, F. H.; Gyawali, S.; Johnston, B.; Filingier, W. (2021) **Presentation and Communication Skills**. *Students@SC Student Programming, Supercomputing 21 Conference, St. Louis, USA*
5. **Dagostini, J.**; Solórzano, A. L.; Girelli, V. (2021) **Por que ainda precisamos falar sobre mulheres em HPC?**. English title: *Why do we still have to talk about women in HPC?* (WHPC Affiliate Brasil - Região Sul) WSCAD Conference, Minas Gerais, Brasil
6. **Dagostini, J.**; Solórzano, A. L.; Girelli, V. (2021) **Latin American Women on HPC: WHPC Affiliate Brasil Região Sul**. *Latin America High Performance Computing Conference (CARLA 2021), Guadalajara, Mexico*
7. **Dagostini, J.**; Solórzano, A. L.; Girelli, V. (2021) **Special Session - Women in High Performance Computing**. (WHPC Affiliate Brasil - Região Sul) XXI Escola Regional de Alto Desempenho, Santa Catarina, Brasil

8. **Dagostini, J.;** Lima, M. V. de M.; Tonin, N. A. (2017) **Enseñar a programar jugando: URI Judge Online.** *WTIC, Resistencia, Argentina*
9. **Dagostini, J.;** Bez, J.L.; Tonin, N. A. (2016) **Portal URI Online Judge - Internacionalização. 2016.** *Tecnomate, Santa Fé, Argentina.*

TUTORIALS AND SHORT TERM COURSES

- May, 2022 **Orquestrando Containers com Docker e LXC**
Cursos de Outono 2022
Short Term Course
- Apr, 2021 **Are you Root? Experimentos Reprodutível em Espaço de Usuário**
XXI Escola Regional de Alto Desempenho da Região Sul
Short Term Course
- Oct, 2020 **Introdução a Programação Paralela com OpenMP**
I Semana Acadêmica Integrada dos Cursos de Ciência da Computação da URI
Short Term Course

COMMUNITY SERVICES

- 2022 – 2025 **COMMITTEE**
Women in HPC Workshop
2024 and 2025 - Workshop Co-Chair
2022 and 2023 - Submissions Co-Chair
- 2023 **ACM/IEEE SC'23 Supercomputing Conference, Denver, CO (USA)**
Infrastructure - Wayfinding Co-Chair
Artifact Description / Article Evaluation (AD/AE) Appendix Reviewer
Lead Student Volunteers Reviewer
- 2023 **International Conference on Parallel Computing**
Reproducibility Committee
- 2022 **29th IEEE Hot Interconnects Symposium**
Web Chair
- 2019 **X Simpósio de Tecnologia da Informação da Região Norte e Noroeste do Rio Grande do Sul**
XXIII Semana Acadêmica do Curso de Ciência da Computação da URI Erechim, Brasil.
Member of the Organizer Team
- 2019 **IX Simpósio de Tecnologia da Informação da Região Norte e Noroeste do Rio Grande do Sul**
XXII Semana Acadêmica do Curso de Ciência da Computação da URI Erechim, Brasil.
Member of the Organizer Team
- 2019 **III Engenharia em 24 Horas: o desafio. (URI Erechim, Brasil)**
Member of the Organizer Team
- EXTERNAL REVIEWER**
- 2022 – Present **Parallel Computing Journal**

VOLUNTEERING

Nov,2020 – Nov,2025

ACM/IEEE SC Conference Series

- SC'25 Selected Lead Student Volunteer (Students@SC HQ)
- SC'24 Selected Lead Student Volunteer (Students@SC Networking)
- SC'22 Selected SCALE/Lead Student Volunteer (Wayfinding Committee)
- SC'21 Selected SCALE/Lead Student Volunteer (Communication Committee)
- SC'20 Selected Regular Student Volunteer

Jan,2021 – Dec, 2022

Women in HPC Affiliate Brasil - Região Sul

Co-Founder of the Women in HPC Affiliate Brasil - Região Sul.

First representation of WHPC Global in Latin America.

Jan,2022 – Dec,2022

BRASA PRÉ Pós-Graduação Mentor

Mentor of a Brazilian student for the application process for graduate schools in USA.

Apr,2022

Volunteer Staff

2022 International Collegiate Programming Contest (ICPC)

South America/Brazil Finals

Principal Activities:

- Help with general organization and coordination in the whole final contest event.

May,2015 – Aug,2017

Volunteer Member

Rotaract Club Erechim Paiol Grande - Rotaract Club recognized by the Rotary International

SKILLS

ENVIRONMENTS

- Working proficiency: Linux/Unix | nano | bash | git | Docker and LXC containers | LaTeX;
- Basic proficiency: spack | slurm | vi.

PROGRAMMING LANGUAGES

- Working proficiency: C | C++ | Python | R | PHP | HTML/CSS;
- Basic proficiency: Java | JavaScript | Ruby.

APIs AND FRAMEWORKS

- Working proficiency: MPI | OpenMP | CakePHP | Score-P;
- Basic proficiency: CUDA | POSIX Threads | OpenACC.

GENERAL SKILLS

- Abilities with leadership and teamwork;
- Experience with teaching algorithms to new programmers;
- Knowledge of teaching didactics (High School with disciplines for teaching degree).